MINI PROJECT

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CLASS:ICSE-B

SUBJECT: DATA STRUCTURE

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1. Aim

To design and implement a ticket booking system that checks seat availability dynamically using **Binary Search**, and performs efficient **sorting and searching operations**.

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2. Problem Statement

A ticket booking system needs to manage seat availability efficiently. The system should store seat numbers, sort them, and allow users to search for a specific seat using binary search to quickly determine availability.

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3. Algorithm

Step 1: Start

Step 2: Input number of seats (n)

Step 3: Enter seat numbers

Step 4: Sort the seat numbers (Ascending)

Step 5: Input seat number to search

Step 6: Apply Binary Search:

- Set low = 0, high = n-1
- Repeat until low <= high
    - mid = (low + high) / 2
    - If seat[mid] == key → Seat Found
    - If key < seat[mid] → Search left half
    - Else → Search right half

Step 7: Display result (Available / Not Available)

Step 8: Stop

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4. C Program

```
import java.util.Scanner;

class TicketSystem {

   public static int binarySearch(int seats[], int key) {

      int low = 0;
```

```
    int high = seats.length - 1;

    while (low <= high) {
        int mid = (low + high) / 2;

        if (seats[mid] == key)
            return mid;
        else if (seats[mid] < key)
            low = mid + 1;
        else
            high = mid - 1;
    }
    return -1; // Seat not found
}

public static void displaySeats(int seats[]) {
    System.out.println("\nAvailable Seats:");
    for (int seat : seats) {
        if (seat != -1)
            System.out.print(seat + " ");
    }
    System.out.println();
}

public static void main(String[] args) {

    Scanner sc = new Scanner(System.in);

    int seats[] = {1,2,3,4,5,6,7,8,9,10};

    int choice, seatNo;

    do {
        System.out.println("\n===== Ticket Booking System =====");
```

```
System.out.println("1. View Available Seats");
System.out.println("2. Book Ticket");
System.out.println("3. Search Seat");
System.out.println("4. Exit");
System.out.print("Enter choice: ");
choice = sc.nextInt();

switch(choice) {

  case 1:
    displaySeats(seats);
    break;

  case 2:
    System.out.print("Enter seat number to book: ");
    seatNo = sc.nextInt();

    int index = binarySearch(seats, seatNo);

    if(index != -1 && seats[index] != -1) {
      seats[index] = -1; // Mark booked
      System.out.println("seat " + seatNo + " booked successfully!");
    } else {
      System.out.println("seat not available!");
    }
    break;

  case 3:
    System.out.print("Enter seat number to search: ");
    seatNo = sc.nextInt();
    index = binarySearch(seats, seatNo);

    if(index != -1 && seats[index] != -1)
      System.out.println("seat Available");
```

```
                else
                    System.out.println("seat Not Available");
                break;

            case 4:
                System.out.println("Thank You!");
                break;

            default:
                System.out.println("Invalid choice!");
        }

    } while(choice != 4);

    sc.close();
  }
}
```

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5. Output:

```
===== Ticket Booking System =====
1. View Available Seats
2. Book Ticket
3. Search Seat
4. Exit
Enter choice: 1

Available Seats:
1 2 3 4 5 6 7 8 9 10

===== Ticket Booking System =====
1. View Available Seats
2. Book Ticket
3. Search Seat
4. Exit
```

```
Enter choice: 2
Enter seat number to book: 5
seat 5 booked successfully!

===== Ticket Booking System =====
1. View Available Seats
2. Book Ticket
3. Search Seat
4. Exit
Enter choice: 1

Available Seats:
1 2 3 4 6 7 8 9 10

===== Ticket Booking System =====
1. View Available Seats
2. Book Ticket
3. Search Seat
4. Exit
Enter choice: 3
Enter seat number to search: 5
seat Not Available

===== Ticket Booking System =====
1. View Available Seats
2. Book Ticket
3. Search Seat
4. Exit
Enter choice: 4
Thank You!
```